

$$\begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$x^2 + 3xy + y^2 = 0$$

$$\frac{1}{4} x'^2 + 3 \cdot \left(-\frac{1}{2} x'\right) \left(-\frac{1}{2} y'\right) + \left(-\frac{1}{2} y'\right)^2 = 0$$

$$\frac{1}{4} x'^2 + \frac{3}{4} x'y' - \frac{1}{4} y'^2 = 0$$

$$x' = -2x \Rightarrow x = -\frac{1}{2} x'$$

$$y' = -2y \Rightarrow y = -\frac{1}{2} y'$$

$$x'^2 + 3x'y' + y'^2 = 0$$

Ellipse

circle

$$\frac{(x-x_1)^2}{a^2} + \frac{(y-y_1)^2}{b^2} = 1$$

$$K = -1$$

$$K = 2$$

$$K = -\frac{1}{2}$$

$$K = -2$$

$$(x-x_1)^2 + (y-y_1)^2 = r^2$$

$$(x, y) \Rightarrow (Kx, Ky)$$

Stretch

① a) Along x-axis ($y=0$); perpendicular to y-axis

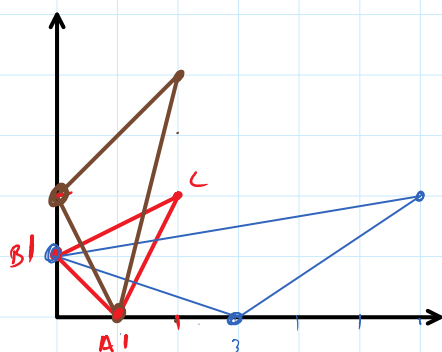
c) with Invariant line y-axis ($x=0$)

$$\begin{pmatrix} K_1 & 0 \\ 0 & 1 \end{pmatrix}$$

Co-ordinates

$$(x, y) \xrightarrow{K_1} (K_1 x, y)$$

A(1, 0)
B(0, 1)
C(2, 2)



S.F. = 3
A'(3, 0)
B'(0, 1)
C'(6, 2)

S.F. = 2
A''(1, 0)
B''(0, 2)
C''(2, 4)

Along y-axis ($x=0$); perpendicular to x-axis ($y=0$)
Invariant line x-axis

$$\begin{pmatrix} 1 & 0 \\ 0 & K_2 \end{pmatrix}$$

$$(x, y) \xrightarrow{K_2} (x, K_2 y)$$

Two way stretching

Two way stretching

$$\begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix} \quad \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$k_1 = k_2 = k \quad \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \Rightarrow \text{scaling}$$

Example

$$(x-2)^2 + (y+1)^2 = 3^2$$

Stretch: Along x-axis $s.f = 2$

$$(2, -1)$$

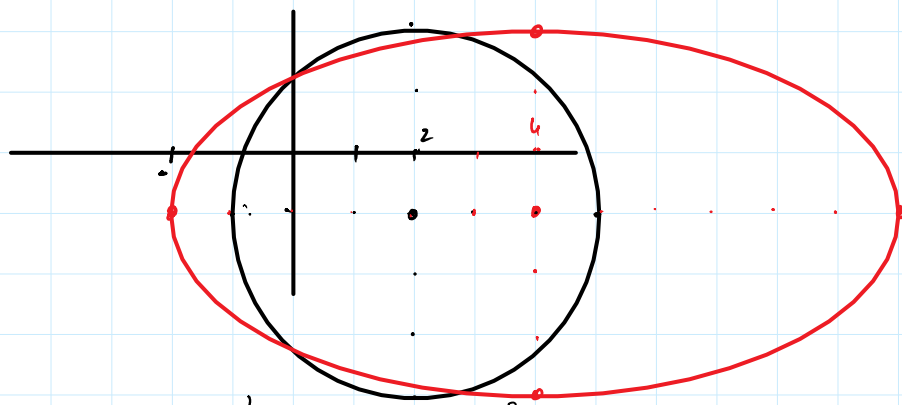
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$x' = 2x$$

$$y' = y$$

$$\Rightarrow \frac{x'}{2} = x$$

$$\frac{(x'-4)^2}{6^2} + \frac{(y'+1)^2}{3^2} = 1$$



$$\left(\frac{x'}{2} - 2\right)^2 + (y' + 1)^2 = 3^2$$

$$\frac{(x' - 4)^2}{4} + \frac{(y' + 1)^2}{1} = 3^2$$

$$\frac{(x' - 4)^2}{4 \times 3^2} + \frac{(y' + 1)^2}{3^2} = 1$$

① $(x, y) \xrightarrow{M} (y, x)$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$(x, y) \xrightarrow{\text{Scale}} (kx, ky)$

$$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$$

$(-2, 3) \xrightarrow{y=x} (3, -2) \xrightarrow{k=2} (6, -4)$

Single matrix $(-2, 3) \rightarrow (6, -4)$ $EM(-2, 3)$

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

$$(4, 5) \rightarrow (6, 8)$$

$$(12, 3) \rightarrow (4, -5)$$

Find a Transformation matrix

$$T A = B$$

$$(12, 3) \rightarrow (4, -5)$$

matrix

$$TA = B$$

$$T \begin{pmatrix} 4 & 12 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 4 \\ 8 & -5 \end{pmatrix}$$

$$T = BA^{-1}$$

$$|A| = 12 - 10 = 2 \quad \text{adj} = \begin{pmatrix} 3 & -2 \\ -5 & 4 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 6 & 4 \\ 8 & -5 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ -5 & 4 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -2 & 4 \\ 49 & -36 \end{pmatrix} \\ = \begin{pmatrix} -1 & 2 \\ 24.5 & -18 \end{pmatrix}$$

Q of $(4, 6) \rightarrow (-2, 6)$

$(8, 4) \rightarrow (-4, 4)$

$$T \begin{pmatrix} 4 & 8 \\ 6 & 4 \end{pmatrix} = \begin{pmatrix} -2 & -4 \\ 6 & 4 \end{pmatrix}$$

find the Transform
& describe
 $\begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$

$$|A| = -32$$

$$\text{adj} = \begin{pmatrix} 4 & -8 \\ -6 & 4 \end{pmatrix}$$

$$T = -\frac{1}{32} \begin{pmatrix} -2 & -4 \\ 6 & 4 \end{pmatrix} \begin{pmatrix} 4 & -8 \\ -6 & 4 \end{pmatrix} = \begin{pmatrix} -8+24 & 16-16 \\ 24-24 & -48+16 \end{pmatrix} \\ = \begin{pmatrix} 16 & 0 \\ 0 & -32 \end{pmatrix} \\ = \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$$

SHEAR

Along x-axis

(Invariant line)

$$T \rightarrow H(1) =$$

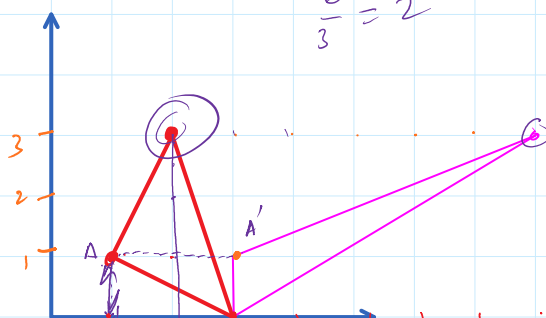
$$(1, 1) \rightarrow (3, 1)$$

$$(2, 3) \rightarrow (8, 3)$$

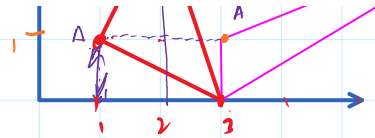
Square → Rhombus

$$\frac{dOIL}{dOIL} = \frac{2}{1} = 2$$

$$\frac{6}{3} = 2$$



$$\begin{aligned} (1,1) &\longrightarrow (3,1) \\ (2,3) &\longrightarrow (8,3) \\ (3,0) &\longrightarrow (3,0) \end{aligned}$$



$$1 + 2(1) = 3$$

$$2 + 2(3) = 8$$

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2+6 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \end{pmatrix}$$

SHEARING Along x-axis

$$\begin{pmatrix} 1 & H_x \\ 0 & 1 \end{pmatrix}$$

Shearing Along y-axis

$$\begin{pmatrix} 1 & 0 \\ H_y & 1 \end{pmatrix}$$

Two way Shearing

$$\begin{pmatrix} 1 & H_x \\ H_y & 1 \end{pmatrix}$$

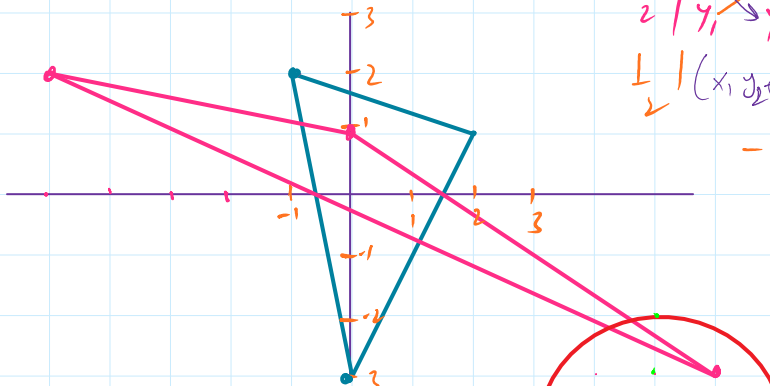
$$A(-1, 2)$$

$$B(0, -3)$$

$$C(2, 1)$$

x-axis Shearing -2

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 2 \\ 2 & -3 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 6 & 0 \\ 2 & -3 & 1 \end{pmatrix}$$

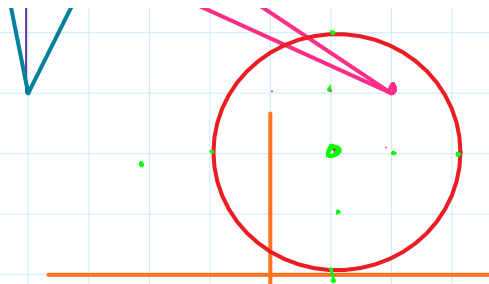


$$\frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

$$\frac{1}{2} (x_1 y_2 + x_2 y_3 + x_3 y_1)$$

$$- (y_1 x_2 + y_2 x_3 + y_3 x_1)$$

$(x-1)^2 + (y-2)^2 = 2^2$
 Shear along y-axis
 shear factor 2



$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$x' = x, \quad y' = 2x + y$$

$$y = y' - 2x$$

$$y = y' - 2x'$$

$$x = x'$$

$$(x'-1)^2 + (y'-2x'-2)^2 = 2^2$$

$$(x'-1)^2 + (\underbrace{y'-2}_{-2x'} - 2)^2 = 2^2$$

$$x'^2 + 1 - 2x' + (y'^2 + 4 + 4x'^2 - 4y' + 8x' - 4x'y') - 4 = 0$$

$$= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$5x'^2 - 4x'y' + y'^2 + 6x' - 4y' + 1 = 0$$

↓

$$A=5, \quad B=-4, \quad C=1 \quad \text{😊}$$

$$B^2 - 4AC = (-4)^2 - 4(5)(1)$$

$$= 16 - 20$$

$$= -4 < 0 \quad \text{ellipse}$$



Quiz #2

Affine Transform

14-May-2026

دائرة 12:00